

Gazebo Multicopter Motor Model Calibration Guide

1. The plugin does not use KV directly

The plugin does **not** accept motor KV directly. It models:

- Thrust: $T = kT \cdot \omega^2$
- Reaction torque: $Q = T \cdot kM$

kT [N/(rad/s)²]

kM [m]

Commands are rotor speeds in **rad/s**.

2. Converting motor KV to maximum rotor speed

- No-load RPM = KV * Battery Voltage
- Loaded RPM $\approx$ 75-90% of no-load RPM
- $\omega = \text{RPM} \cdot 2\pi/60$
- Use loaded ω for ``.

KV estimates maximum speed only.

3. Finding motorConstant

$kT = \text{Thrust}(\text{Newtons}) / \omega^2$

Convert grams:

$\text{Thrust}(\text{N}) = \text{grams}/1000 \cdot 9.80665$

Fit multiple measurements.

4. Finding momentConstant

$\text{momentConstant} = \text{Torque} / \text{Thrust}$

Typical values: 0.01-0.03 m.

5. Configure each motor

Each rotor needs:

- jointName
- linkName
- actuator_number
- turningDirection
- maxRotVelocity
- motorConstant
- momentConstant
- timeConstantUp
- timeConstantDown
- rotorDragCoefficient

- rollingMomentCoefficient

6. Motor response time constants

Use first-order dynamics:

- timeConstantUp
- timeConstantDown

Estimate tau from step response.

7. rotorVelocitySlowdownSim

Not a physical parameter.

Usually keep 10.

8. Rotor drag coefficients

Tune rotorDragCoefficient and rollingMomentCoefficient after thrust is correct.

9. Verify hover speed

$$\omega_{\text{hover}} = \sqrt{(m \cdot g) / (4 \cdot kT)}$$

Hover speed should be well below maxRotVelocity.

10. Sending commands

The plugin expects rotor speed in rad/s.

11. Recommended calibration order

1. Mass and inertia
2. Rotor geometry
3. maxRotVelocity
4. motorConstant
5. momentConstant
6. Time constants
7. Drag coefficients
8. Hover verification
9. Flight testing

Key takeaway

KV determines achievable rotor speed. Thrust stand data determines motorConstant.